SU Youyou

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EDUCATION

NANYANG TECHNOLOGICAL UNIVERSITY (NTU), SINGAPORE Singapore
Bachelor of Science in Physics and Mathematical Sciences Expected Jun 2028
GPA: 4.85/5.00
Relevant Coursework: Graduate Quantum Information, Quantum Mechanics, Differential Geometry, ODE, Topology, Real Analysis, Linear Algebra, Calculus, Python, R Code

HONORS & AWARDS

NANYANG TECHNOLOGICAL UNIVERSITY
- Dean's List, AY2025/2026 (Top 5% of the cohort)
- NTU President Research Scholar with Merit, AY2025/2026
- NTU Science and Technology Undergraduate Scholarship (full ride)

EXPERIENCE

URECA 2026-27 (ONGOING) NTU Singapore
Undergraduate Researcher Aug 2026 - Present
Supervisor: Prof. Gu Mile, Dr. Graeme Berk
- Studied computational mechanics and information-theoretic measures of memory and learning in stochastic processes
- Investigated the operational meaning of transient information as a measure of the information required during the learning process
- Analyzed a process with almost no internal structure that can nevertheless be arbitrarily difficult to learn from observations

TSUNG-DAO LEE INSTITUTE, SHANGHAI JIAO TONG UNIVERSITY Shanghai
Research Assistant May 2026 - Jul 2026
Supervisor: Assoc. Prof. Liu Liang
- Studied Abrikosov vortex dynamics, including overdamped motion, vortex-vortex interactions, pinning, and thermal driving, with potential relevance to vortex-based manipulation of Majorana zero modes (MZMs) in topological quantum computing
- Developed a two-dimensional numerical model for vortex manipulation under localized electrothermal heating
- Simulated vortex capture, transfer, and controlled trajectories, and presented the findings in a research poster

EASYCV – FULL-STACK RESUME BUILDER Online
Independent Web Developer Jun 2026
- Designed and built a full-stack resume builder using Next.js and TypeScript, enabling users to generate professional, ATS-friendly resumes from structured input
- Integrated a LaTeX-based PDF generation pipeline with live preview, multilingual support, and automated document compilation
- Containerized and deployed the application using Docker and Railway; built and generated this resume using the platform itself
- Continuously iterated on the platform through hands-on use, adding browser autosave, private cross-device cloud backup, and responsive UI improvements
- Live Demo: https://resume-builder-demo-production.up.railway.app
- GitHub: https://github.com/suyouyoualan-sketch/resume-builder-demo

URECA 2025-26 & ODYSSEY SUMMER RESEARCH NTU Singapore
Undergraduate Researcher Aug 2025 - Jul 2026
Supervisor: Assoc. Prof. François Gay-Balmaz
- Studied quantum mechanics from first-principles mathematical foundations, emphasizing rigorous derivations
- Covered Hilbert spaces, quantum states and operators, unitary dynamics, and the Schrödinger, Heisenberg, and path integral pictures
- Extended the study to multiparticle systems, Fock spaces, second quantization, and creation and annihilation operators
- Completed two individual-study papers on these topics

LEADERSHIP & ACTIVITIES

SPMS LEARNING CENTER Singapore
Peer Tutor Aug 2026 - Present
- Tutor junior students on analytical mechanics weekly

NTU FENCING Singapore
Member (Épée) Aug 2026 - Present
• Participate in weekly épée training sessions, focusing on technical skills, footwork, and competitive fencing techniques

SKILLS & INTERESTS

Technical: Python, R Code, LaTeX, PennyLane
Language: Mandarin native, English fluent
Interests: Piano, Fencing, Ping Pong, Korean Language